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Reactivities of mixed organozinc and mixed organocopper reagents: 9. Solvent dependence of group transfer selectivity in sp^3C coupling and acylation of mixed diorganocuprates and diorganozincs

Ender Erdik [a,*], Fatma Eroğlu [a], Melike Kalkan [a], Özgen Ömür Pekel [a], Duygu Özkan [a], Ebru Z. Serdar [b]

[a] Ankara University, Science Faculty, Beşevler, Ankara 06100, Turkey
[b] Dermikaya Center, Kadıköy, İstanbul, Turkey

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ABSTRACT

The selectivity and/or reactivity of organyl group transfer of mixed diorganocuprates in their alkyl coupling in THF depends on N- or O-donor solvents as cosolvents. Selective *n*-Bu group transfer is observed in room temperature alkylation of Grignard reagent derived stoichiometric *n*-BuPhCuMgBr reagent in THF:co-solvent and solvation effects do not change the group transfer ability. However, in the alkylation of catalytic mixed cuprates derived from CuI catalyzed *n*-$BuPh_2ZnMgBr$ and *n*-$Bu_2PhZnMgBr$, group transfer ability depends on the solvation effect and it can be controlled by using N- or O-donor solvents. In alkylation of CuI catalyzed mixed zincate *n*-$BuPh_2ZnMgBr$ and also *n*-$Bu_2PhZnMgBr$ in THF at reflux temperature Ph group transfer takes place (*n*-Bu/Ph transfer ratio is 1:9 and 4:6, respectively) whereas *n*-Bu transfer increases in THF:NMP (1:1) resulting *n*-Bu/Ph transfer ratio of 4:6 and 8:2, respectively. Group transfer ability in allylation of *n*-BuPhZn seems not to be solvent dependent. The solvent effect on the group transfer ability has been found to be dependent also on the R^1 and R^2 partnership in room temperature benzoylation of catalytic mixed cuprates, R^1R^2CuZnI, derived from CuI catalyzed R^1R^2Zn. These results are briefly discussed in terms of solvation of mixed diorganocuprate and diorganozinc reagents and provide useful information in their atom-economic alkyl, allyl and acyl coupling reactions.

1. Introduction

Diorganocuprates R_2CuM (M = Li, MgX) have provided one of the most consistently popular organometallic reagents for C–C bond formation [1–3]. However, the synthetic problem associated with the use of copper nucleophile, $R_2Cu^{\ominus}$ is that only one of the R groups can be transferred to the electrophile. Organozinc reagents are the other most used organometallic reagents due to their compatibility of many functional groups. However, in the reactions of diorganozincs, R_2Zn [4–7], and triorganozincates, R_3ZnM (M = Li, Na, MgX) [7,8], which are more reactive than organozinc halides, RZnX [4–7], the problem of one R group transfer is again a serious limitation to the atom-economic use of diorganozincs and triorganozincates. The problem has been solved by (i) preparing mixed organozinc and copper reagents of R^1R^2CuM [9], R^1R^2Zn [10], and $R_1R^2_2ZnM$ type [11], in which the R^2 group has a lower transfer rate than R^1 group and also by (ii) preparing mixed organozinc and copper reagents of R_RR_TCuM [9,12], R_RR_TZn [10,13] and $(R_R)_2R_TZnM$ type [11,14] composed of one transferable group R_T together with the residual group, R_R. A variety of mixed diorganocuprates and mixed diorganozincs have found extensive application in organometallic synthesis. Mixed triorganozincates have not been used extensively. However, recently (iii) catalytic mixed triorganozincates have been prepared using R^1MgBr and catalytic amounts of $ZnCl_2$ [7,15a–d] and R^2_2Zn containing residual R^2 groups, respectively [15a,e,f].

We are currently working on the mechanisms and controlling factors of group transfer selectivity in the reactions of mixed diorganocuprates [16,17], diorganozincs [18–22], and triorganozincates [23] and their synthetic applicability [19–21].

So far, we observed that the group transfer selectivity in mixed diorganocuprates and diorganozincs depend not only on the reaction mechanism, but also (i) on the counter ion of the cuprates R^1R^2CuM (M = MgX, ZnX) and (ii) on the reaction parameters, i.e. solvent, temperature and reaction time. We recently reported that the transfer ability of organyl groups in *n*-BuPhZn in their Cu(I) catalyzed reaction with benzoyl chloride in THF can be controlled using N- or O-donor solvents [18]. *n*-Bu group:Ph group transfer ratio is 6:4 in THF and 9:1 in THF:NMP (2:1), but 1:9 in THF:TMEDA

* Corresponding author. Ankara Üniversitesi, Fen Fakültesi, Kimya Bölümü, Beşevler, Ankara 06100, Turkey. Tel.: +90 312 212 67 20x1039; fax: +90 312 223 23 95.
E-mail address: erdik@science.ankara.edu.tr (E. Erdik).

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(2:1). Then, we carried out a brief study to gain some qualitative insight for the solvent effect on the group transfer ability of mixed diorganocuprates and also mixed diorganozincs in their sp^3C coupling reactions. In this paper, we wish to report the performance of donor solvents with the aim of finding the best solvent

(i) in the alkylation of stoichiometric mixed diorganocuprate, *n*-BuPhCuMgBr,
(ii) in the alkylation of catalytic mixed diorganocuprate, *n*-BuPhCuMgBr derived from zincates *n*-Bu_xPh_{3-x}ZnMgBr ($x = 1$ and 2) in the presence of CuI catalysis and
(iii) in the allylation of mixed diorganozinc, *n*-BuPhZn,

and we give a short explanation for the solvent effect on their group transfer ability.

We already gave a brief explanation for the solvent effect in acylation of catalytic *n*-BuPhCuZnI, derived from *n*-BuPhZn in the presence of CuI catalysis [18]. However, from these results, we raised the question of how the solvent affects the group selectivity in the acylation of other catalytic mixed diorganocuprates. So, we also examined

(iv) the performance of THF and THF:TMEDA (1:2) as solvents in the acylation of a number of catalytic mixed diorganocuprates, R^1R^2CuZnI derived from mixed diorganozincs, R^1R^2Zn in the presence of CuI catalysis.

The results may provide useful information when mixed diorganocuprates and diorganozincs are used for atom-economic synthesis.

2. Results and discussion

For the solvent effect on the group transfer ability of mixed diorganocuprate and diorganozinc reagents in their sp^3C coupling reactions, we firstly focused on the sp^3C alkyl coupling reactions of mixed bromomagnesium diorganocuprates and secondly allyl coupling reactions of mixed diorganozincs planning to see also the dependence of solvent effect on the reaction mechanism. In acylation reactions, we screened the solvent effect on the group transfer ability on a number of mixed cuprates using THF and THF:TMEDA (1:2) as selectivity controlling solvents and/or solvent mixtures.

As model reactions, we chose (i) coupling of stoichiometric *n*-BuPhCuMgBr (**1ab**) reagent (Scheme 1a) and (ii) coupling of catalytic *n*-BuPhCuMgBr reagents formed *in situ* from *n*-$BuPh_2$ZnMgBr (**2ab₂**) and *n*-Bu_2PhZnMgBr (**2a₂b**) in the presence of CuI catalyst, i.e. catalytic *n*-BuPhCuMgBr reagents (Scheme 1b) with *n*-pentyl bromide **3** as alkyl halide. For diorganozincs, which do not couple with alkyl halides, we investigated (iii) coupling of *n*-BuPhZn (**6ab**) with allyl bromide (Scheme 1c). As cosolvents, we used N-donor and O-donor solvents and also toluene as a noncoordinating solvent for comparison and we found the coupling yield and relative transfer ability *n*-Bu and Ph groups in THF:cosolvent in competition with the outcome of the reaction in THF alone. (iv) For acylation reactions, catalytic mixed diorganocuprates containing alkyl, benzyl, aryl and alkenyl groups were prepared *in situ* using mixed diorganozincs and CuI catalysis and they were allowed to react with benzoyl chloride in THF and in THF:TMEDA (1:2) (Scheme 1d).

2.1. Coupling of n-BuPhCuMgBr with n-pentyl bromide

Optimized yields and **4**:**5** product ratios for the alkylation of *n*-BuPhCuMgBr in THF and in THF:cosolvent at room temperature (Scheme 1a) are summarized in Table 1. Coupling yields in THF:cosolvent (1:1) mixtures and background yields for alkylation of homocuprates in THF were already found [16].

As seen, the total yield and **4**:**5** product ratio (98:2–90:10) did not change much in THF:NMP (1:1), in THF:DMPU (1:1) and in THF:diglyme (1:1) (entries 2, 4 and 6) compared to the reaction in THF (entry 1). Using THF:HMPA (1:1) as solvent decreased the yield (entry 7) and THF:TMEDA (1:1) resulted in a much lower yield (entry 9). However, selective *n*-Bu group transfer takes place in the presence of coordinating solvents. Using toluene as a cosolvent decreased the yield to 23%, but the product ratio remained the same. Carrying out the coupling in THF:cosolvent (1:2) (entries 5, 8 and 10) resulted in about 30% decrease in the yield in all cases, however did not make a remarkable change in the product ratio.

We can see that the use of a R^1R^2CuMgBr ($R^1 = n$-alkyl, $R^2 =$ aryl) instead of R^1_2CuMgBr for alkyl coupling in THF increases *n*-alkyl transfer (in the case of *n*-BuPhCuMgBr, from 71% to 90%); however using a N- or O-donor solvent in 1:1 molar ratio or less does not change either the yield or the relative group transfer ability.

2.2. Coupling of CuI catalyzed n-BuPh₂ZnMgBr and n-Bu₂PhZnMgBr with n-pentyl bromide

As we already observed that CuI catalyzed alkylation of *n*-$BuPh_2$ZnMgBr (**2ab₂**) and *n*-Bu_2PhZnMgBr (**2a₂b**) in THF lead to higher yields at reflux temperature [23], we investigated the group

$$\underset{\textbf{1ab}}{n\text{-BuPhCuMgBr}} + \underset{\textbf{3}}{n\text{-PentBr}} \xrightarrow[\text{THF:cosolvent, } 25\ ^\circ\text{C, 3h}]{} \underset{\textbf{4}}{n\text{-}C_9H_{20}} + \underset{\textbf{5}}{n\text{-PentPh}} \quad \text{(a)}$$

$$\underset{\textbf{2}}{n\text{-}Bu_xPh_{3\text{-}x}\text{ZnMgBr}} + \underset{\textbf{3}}{n\text{-PentBr}} \xrightarrow[\text{THF:cosolvent, } 65\ ^\circ\text{C, 2h}]{20\text{ mol\% CuI}} \underset{\textbf{4}}{n\text{-}C_9H_{20}} + \underset{\textbf{5}}{n\text{-PentPh}} \quad \text{(b)}$$

x = 0-3

$$\underset{\textbf{6ab}}{n\text{-BuPhZn}} + \underset{\textbf{7}}{H_2C{=}CHCH_2Br} \xrightarrow[\text{THF:cosolvent, } 25\ ^\circ\text{C, 30 min}]{} \underset{\textbf{8}}{H_2C{=}CHCH_2(n\text{-Bu})} + \underset{\textbf{9}}{H_2C{=}CHCH_2Ph} \quad \text{(c)}$$

$$\underset{\textbf{6}}{R^1R^2Zn} + \underset{\textbf{10}}{PhCOCl} \xrightarrow[\text{THF or THF:TMEDA (1:2), } 25\ ^\circ\text{C, 20 min}]{10\text{ mol\% CuI}} \underset{\textbf{11}}{R^1COPh} + \underset{\textbf{12}}{R^2COPh} \quad \text{(d)}$$

Scheme 1.

Table 1
Effect of cosolvents on the organic group transfer ability in the reaction of bromomagnesium *n*-butylphenylcuprate **1ab** with *n*-pentyl bromide in THF at room temperature.[a,b]

$$\underset{\mathbf{1ab}}{n\text{-BuPhCuMgBr}} + \underset{\mathbf{3}}{n\text{-PentBr}} \xrightarrow[25\,^{\circ}\text{C},\ 3\text{h}]{} \underset{\mathbf{4}}{n\text{-C}_9\text{H}_{20}} + \underset{\mathbf{5}}{n\text{-PentPh}}$$

Entry	Solvent	Total yield, %[c] (**4**:**5**)[d]
1	THF	93 (98:2)
2	THF:NMP (1:1)	98 (90:10)
3	THF:NMP (1:2)	65 (89:11)
4	THF:DMPU (1:1)	92 (91:9)
5	THF:DMPU (1:2)	63 (84:16)
6	THF:diglyme (1:1)	84 (95:5)
7	THF:HMPA (1:1)	73 (95:5)
8	THF:HMPA (1:2)	46 (98:2)
9	THF:TMEDA (1:1)	21 (90:10)
10	THF:TMEDA (1:2)	6 (83:17)
11	THF:toluene (1:1)	23 (95:5)
12	THF:toluene (1:2)	9 (95:5)

[a] Molar ratio of **1ab**:**3** was used to be 3:1.
[b] GC yields for alkylation of n-$Bu_2CuMgBr$ $\mathbf{1a_2}$ and $Ph_2CuMgBr$ $\mathbf{1b_2}$ are 71% and 8%, respectively [16].
[c] The sum of GC yields of **4** and **5**.
[d] The ratio of GC yields of **4** and **5**.

transfer ability of organic groups in THF:cosolvent at this temperature. Optimized yields and **4**:**5** product ratios are given in Table 2. To see the effect of solvent on the performance of n-$BuPh_2ZnMgBr$ ($\mathbf{2ab_2}$)/"CuI" and n-$Bu_2PhZnMgBr$ ($\mathbf{2a_2b}$)/"CuI" we carried out studies in parallel with homozincates n-$Bu_3ZnMgBr$ ($\mathbf{2a_3}$)/"CuI" and $Ph_3ZnMgBr$ ($\mathbf{2b_3}$)/"CuI".

As seen in the coupling of n-$BuPh_2ZnMgBr$/"CuI", transfer of only Ph group takes place in THF (**4**:**5** ratio is 8:92) (entry 1). However, transfer ability of *n*-Bu group appreciably increases in THF:NMP (1:1) (**4**:**5** ratio is 44:56) (entry 2) and in THF:DMPU (1:1) (**4**:**5** ratio is 43:57) (entry 3). In THF:diglyme (1:1) (entry 4) and in THF:TMEDA (entry 5), the coupling again results in selective Ph group transfer (**4**:**5** ratios are 14:86 and 20:80, respectively). So, it seems that NMP and DMPU as a cosolvent have an effect to increase the relative transfer ability of alkyl groups in the CuI catalyzed alkylation of $R^1R^2{}_2ZnMgBr$ (R^1 = *n*-alkyl, R^2 = aryl) reagents.

In the coupling of n-$Bu_2PhZnMgBr$/"CuI", the results are similar to those obtained with n-$Bu_2PhZnMgBr$/"CuI". Transfer of mostly Ph group takes place in THF (**4**:**5** ratio is 38:62) (entry 1) and selective *n*-Bu group transfer takes place in THF:NMP (1:1) (**4**:**5** ratio is 98:2) (entry 2) and in THF:DMPU (1:1) (**4**:**5** ratio is 98:2) (entry 3). Coupling in THF:TMEDA (1:1) also resulted in selective *n*-Bu group transfer (**4**:**5** ratio is 81:19) (entry 5). We were delighted to see that organic group transfer in CuI catalyzed *n*-alkyl coupling $R^1{}_2R^2ZnMgBr$ (R^1 = *n*-alkyl, R^2 = aryl) reagents can be controlled by using N- or O-donor cosolvents in THF.

In controlling the background yields, we observed lower coupling yield for CuI catalyzed homozincate n-$Bu_3ZnMgBr$ in the presence of diglyme and TMEDA as cosolvents (entries 4 and 5, respectively) and for CuI catalyzed homozincate $Ph_3ZnMgBr$ in the presence of DMPU and diglyme as cosolvents (entries 3 and 4, respectively) compared to the yields obtained in THF. Then we expected to see a similar decrease for *n*-Bu group transfer in the presence of diglyme and/or TMEDA and a decrease for Ph group transfer in the presence of DMPU and/or diglyme in the CuI catalyzed coupling yields of both mixed zincates. As expected, we observed that the presence of diglyme and TMEDA in coupling of n-$buPh_2ZnMgBr$/"CuI" and diglyme in coupling of n-$Bu_2PhZnMgBr$/"CuI" lead to minimum *n*-Bu group transfer. A decrease in Ph group transfer was also observed in CuI coupling of both zincates, but only in the presence of DMPU. From a comparison of CuI catalyzed coupling yields of homozincates and mixed zincates, the use of NMP, DMPU or especially TMEDA as a cosolvent in THF appears promising in atom-economic *n*-alkyl–*n*-alkyl coupling using $R^1{}_2R^2ZnMgBr$ (R^1 = *n*-alkyl, R^2 = aryl) reagents and *n*-alkyl bromides (entries 2, 3 and 5). It is also apparent that coupling of $R^1R^2{}_2ZnMgBr$ (R^1 = *n*-alkyl, R^2 = aryl) reagents in THF rather than homozincates, $R^2{}_3ZnMgBr$ seems to be the best choice for aryl-*n*-alkyl coupling (entry 1).

In order to propose a brief convincing explanation for the solvent dependence of the relative transfer ability of *n*-Bu and Ph groups in the alkylation of stoichiometric mixed cuprate *n*-BuPhCuMgBr (**1ab**), we assumed a contact ion pair (CIP) such as a dimer **A** or a hetero aggregate **B** in equilibrium with solvent-separated ion pair (SSIP) (Scheme 2) [16,17,23]. The structure of similar to lithium homo and mixed diorganocuprates [2,3,9j,24–29] and Grignard reagent derived homocuprates [24,26–30] were determined by crystallographical and NMR spectral studies. These findings

Table 2
Effect of cosolvents on the organic group transfer ability in the reaction of CuI catalyzed alkylation of bromomagnesium *n*-butyldiphenylzincate $\mathbf{2ab_2}$, di(*n*-butyl) phenylzincate $\mathbf{2a_2b}$, tri-*n*-butylzincate $\mathbf{2a_3}$ and triphenylzincate $\mathbf{2b_3}$ with *n*-pentyl bromide in THF at reflux temperature.[a]

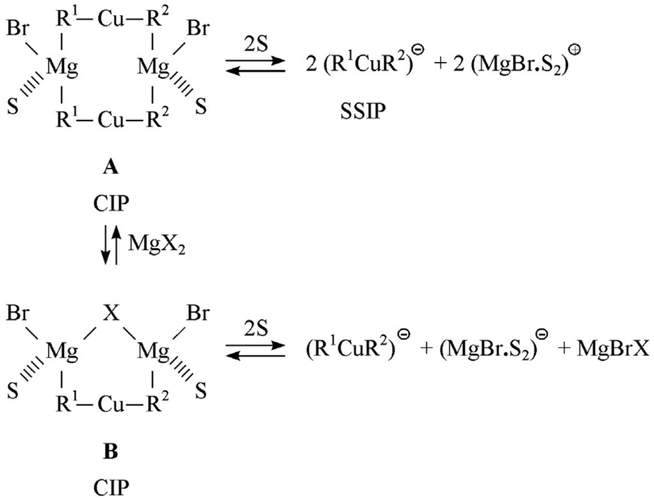
$$\underset{\mathbf{2}}{n\text{-Bu}_x\text{Ph}_{3\text{-}x}\text{ZnMgBr}} + \underset{\mathbf{3}}{n\text{-PentBr}} \xrightarrow[\text{THF:cosolvent, 65 °C, 2h}]{\text{20 mol\% CuI}} \underset{\mathbf{4}}{n\text{-C}_9\text{H}_{20}} + \underset{\mathbf{5}}{n\text{-PentPh}}$$

x=0-3

n-$Bu_3ZnMgBr$ $\mathbf{2a_3}$, $Ph_3ZnMgBr$ $\mathbf{2b_3}$, n-$BuPh_2ZnMgBr$ $\mathbf{2ab_2}$, n-$Bu_2PhZnMgBr$ $\mathbf{2a_2b}$

Entry	Solvent	Yield, %[b]		Total yield, %[c] (**4**:**5**)[d]	
		With $\mathbf{2a_3}$	With $\mathbf{2b_3}$	With $\mathbf{2ab_2}$	With $\mathbf{2a_2b}$
1	THF	78	58	59 (8:92)	61 (38:62)
2	NMP (1:1)	65	48	55 (44:56)	70 (80:20)
3	DMPU (1:1)	70	34	58 (43:57)	54 (98:2)
4	Diglyme (1:1)	39	35	35 (14:86)	30 (28:72)
5	TMEDA (1:1)	24	53	61 (20:80)	77 (81:19)
6	Toluene (1:1)	41	60	46 (9:91)	38 (40:60)

[a] Molar ratio of **2**:**3** was used to be 1.5:1.
[b] GC yield.
[c] The sum of GC yields of **4** and **5**.
[d] The ratio of GC yields of **4** and **5**.



Scheme 2.

indicated the equilibrium between CIP and SSIP structures which are in accordance with reactivities [9j,24,30].

In general, CIP is dominant in weakly solvating solvents and SSIP is preferred in solvents with strong coordination ability [3,24,31,32]. However, it is known that it is essentially only CIP structures of a cuprate which undergoes the reaction and the reactions proceed with a small equilibrium concentration of CIP in solution and SSIP is the much less or even unreactive species [3,9j,26,30,32–37].

Coupling of stoichiometric *n*-BuPhCuMgBr (**1ab**) in THF and in THF:cosolvent (1:1) affords almost the same yield (cosolvent = NMP, DMPU, HMPA and diglyme). However, the yields in THF:cosolvent (1:2) compared to those in THF:cosolvent (1:1) are quite low. These results provide support for the equilibrium in favour of SSIP in strongly coordinating solvents resulting in the predominance of the unreactive SSIP in these solvents.

On the contrary, TMEDA as a cosolvent gave unexpectedly low yield even using THF:TMEDA (1:1) as solvent. We think that TMEDA can afford strong complexation with two $MgBr^{\oplus}$ counter ions as a bridging ligand [26]. In this case, the equilibrium lying completely on the side of SSIP can prevent the presence of CIP structure in a small equilibrium concentration to react. However, quite a low yield for the coupling in THF:toluene (1:1) is not an unexpected behaviour. Since, in this case, cosolvent has no coordination ability, but diminished nucleophilic character of CIP structure will possibly prevent formation of coupled products although the equilibrium is far beyond the CIP structures.

The selective *n*-Bu group transfer in the coupling of *n*-BuPhCuMgBr with 90% yield in THF compared with coupling yields of homocuprates, i.e. 71% yield of *n*-Bu_2CuMgBr (**1a₂**) and 8% yield of Ph_2CuMgBr (**1b₂**) simply shows that the residual group Ph facilitates the transfer of *n*-Bu group [9a].

As active catalytic intermediates in the copper catalyzed reactions of mixed zincates, *n*-BuPh_2ZnMgBr (**2ab₂**) and *n*-Bu_2PhZnMgBr (**2a₂b**), we may assume transmetallation of *n*-butyl and phenyl carbanions which are known to be in equilibrium in small amounts with zincate as Grignard reagents and diorganozincs (1) [8j,23,36–38]. Transmetallation of Grignard reagents possibly result in formation of *n*-BuCu and PhCu and then formation of homo and mixed cuprates. As zinc cuprates do not couple with *n*-pentyl bromide, transmetallation of diorganozincs are omitted.

Mixed zincate *n*-BuPh_2ZnMgBr (**2ab₂**) has the possibility of forming Ph_2CuMgBr (**1b₂**) and *n*-BuPhCuMgBr (**1ab**) more than *n*-Bu_2CuMgBr (**1a₂**) in *in situ* Zn → Cu transmetallation. Similarly, formation of *n*-Bu_2CuMgBr (**1a₂**) and *n*-BuPhCuMgBr (**1ab**) is expected to be more than Ph_2CuMgBr (**1b₂**) in *in situ* Zn → Cu transmetallation of mixed zincate *n*-Bu_2PhZnMgBr (**2a₂b**). These assumptions seem to be in accordance with the group transfer selectivity in the coupling of *n*-BuPh_2ZnMgBr (**2ab₂**) and *n*-Bu_2PhZnMgBr (**2a₂b**) derived catalytic mixed cuprates (Table 2, entry 1).

Our brief study suggests that using a N- or O-donor solvent as a cosolvent does not change *n*-Bu group transfer ability in the alkylation of *n*-BuPhCuMgBr (**1ab**) in THF. However, the group transfer abilities of mixed zincate derived catalytic cuprates in alkyl coupling are solvent dependent. These findings support the difference in the reactivity of Grignard reagent derived and zincate derived diorganocuprate reagents. So, further comments on the group selectivity of mixed zincate derived catalytic cuprates must await more information concerning structure of these reagents.

2.3. Coupling of n-BuPhZn with allyl bromide

We also investigated uncatalyzed coupling of *n*-BuPhZn with allyl bromide at room temperature (Scheme 1c) to see the effect of the reaction mechanism and also N-donor solvent as a cosolvent on the group transfer ability.

Coupling of *n*-BuPhZn takes place by reaction of *n*-Bu group and/or Ph group as nucleophiles with allyl bromide in THF quantitatively without Cu(I) catalysis. Optimized yields and **8**:**9** product ratios for Relative transfer of *n*-Bu and Ph groups are shown in Table 3.

Background yields for coupling of *n*-Bu_2Zn (**6a₂**) and Ph_2Zn (**6b₂**) in THF were found to be 92% and 98%, respectively. The yield remained same in the presence of NMP and HMPA as N-donor cosolvents; NMP as a cosolvent noticeably increased **8**:**9** product ratio (entry 2), but HMPA did not result in a much change (entry 3). It is interesting that the yield decreased much in the presence of TMEDA and selective Ph group transfer took place (entry 4).

According to the literature data, the relative Ph > *n*-Bu transfer in allyl coupling may not be surprising. Since in the Ni catalyzed reaction of mixed diorganozincs with anhydrides, the relative transfer ability of organyl groups was reported to be Ph > Me > Et >> *i*-Pr [39] and theoretical study also supported the higher reactivity of Ph compared to Et group for the reaction of diphenylzinc/diethylzinc with aldehydes [10j]. Increased *n*-Bu coupling in the presence of NMP and HMPA is also in accordance with the smooth allylation of dialkylzincs in the presence of N-donor solvents [40]. Chelating diamines, such as TMEDA were also reported as the key for successful aryl transfer from mixed alkylarylzincs in their reactions with aldehydes [10m]. In the allylation of *n*-BuPhZn, similarly, selective Ph coupling takes place in THF:TMEDA compared to THF alone, but with a quite low yield (entry 4). We also wanted to see how the yield and **8**:**9** product ratio would change using chelating diamines as additives. In the presence of 1 mol equiv of bipyridyl and urotropine, the yield decreased to 85% and 65%, respectively, but **8**:**9** product ratio was found not much different than that in THF alone (entries 5 and 6). Using THF:toluene (1:1) as solvent just decreased the yield as excepted, but did not change **8**:**9** product ratio (entry 7). So, it seems that

$$n\text{–BuPh}_2\text{ZnMgBr} \rightleftarrows n-\text{BuMgBr} + \text{Ph}_2\text{Zn}\ \text{and/or}\ \text{PhMgBr} + n-\text{BuPhZn} \xrightarrow{\text{CuI}} n-\text{BuCu}\ \text{and/or}\ \text{PhCu} \xrightarrow{n\text{–BuPh}_2\text{ZnMgBr}}$$
$$\underset{1a_2}{n-\text{Bu}_2\text{CuMgBr}} + \underset{1b_2}{\text{Ph}_2\text{CuMgBr}} + \underset{1ab}{n-\text{BuPhCuMgBr}} \qquad (1)$$

Table 3
Effect of cosolvents on the organic group transfer ability in the reaction of *n*-butylphenylzinc **6ab** with allyl bromide **7** in THF at room temperature.[a,b]

$$\underset{\textbf{6ab}}{n\text{-BuPhZn}} + \underset{\textbf{7}}{H_2C{=}CHCH_2Br} \xrightarrow[\text{25 °C, 30 min}]{\text{THF:cosolvent,}} \underset{\textbf{8}}{H_2C{=}CHCH_2(n\text{-Bu})} + \underset{\textbf{9}}{H_2C{=}CHCH_2Ph}$$

Entry	Solvent	Total yield, %[c] (**8**:**9**)[d]
1	THF	99 (17:83)
2	THF:NMP (1:1)	100 (40:60)
3	THF:HMPA (1:1)	98 (26:74)
4	THF:TMEDA (1:1)	20 (0:100)
5	Bipyridyl[e]	85 (27:63)
6	Urotropine[e]	67 (31:69)
7	THF:toluene (1:1)	72 (21:79)

[a] Molar ratio of **6ab**:**7** was used to be 1.5:1.
[b] GC yields for allylation of *n*-Bu_2Zn **6a₂** and Ph_2Zn $\mathbf{6b_2}$ are 92% and 98%, respectively.
[c] The sum of GC yields of **8** and **9**.
[d] The ratio of GC yields of **8** and **9**.
[e] 1 mol equiv of additive was used.

using THF alone as a solvent is the best choice for atom-economic aryl-allyl coupling using *n*-alkylarylzincs and allylic bromides.

2.4. Acyl coupling of CuI catalyzed R^1R^2Zn with benzoyl chloride

We already found that the group transfer ability in the CuI catalyzed benzoylation of *n*-BuPhZn in THF at room temperature can be controlled using N- and O-donor cosolvents [18]. For *n*-Bu group selective and Ph group selective acylation in THF and THF:TMEDA (1:2), respectively, we offered an explanation depending on the acylation mechanism of solvent coordinated cuprates.

In order to observe if the group selectivity control of THF and THF:TMEDA (1:2) works for other mixed diorganozincs, R^1R^2Zn, **6** in their CuI catalyzed acylation, we determined the optimized yields and **11**:**12** product ratios (Table 4).

As seen, CuI catalyzed acylation of Ph($PhCH_2$)Zn **6bc** (entry 8) results in $PhCH_2$ selective reaction in THF ($PhCH_2$/Ph transfer ratio is 79:21) and $PhCH_2$ group transfer ability decreases in THF:TMEDA ($PhCH_2$/Ph transfer ratio is 42:58). These results are similar to those obtained in the acylation of *n*-BuPhZn **6ab** (entry 5). In the CuI catalyzed acylation of *n*-Bu($PhCH_2$)Zn **6ac**, the higher transfer ability of $PhCH_2$ group than that of *n*-Bu group both in THF and also in THF:TMEDA seems surprising (entry 6). However, in the acylation of catalytic cuprate derived from Ph(MeCH═CH)Zn **6bd** (entry 9), MeCH═CH group results in a high decrease in the yield of Ph group transfer in THF:TMEDA and Ph group transfer seems not to take place in THF compared to the background yields of Ph_2Zn $\mathbf{6b_2}$ (entry 2). A similar result was obtained in the acylation of catalytic cuprate derived from ($PhCH_2$)(MeCH═CH)Zn **6cd** and quite a low total yield was obtained (entry 10). However, some MeCH═CH group transfer in THF:TMEDA (1:2) seemed interesting implying that in the presence of aryl group as a partner, alkenyl group may be reactive in acylation whereas in the presence of benzyl group it appears inactive. As we could not pre($PhCH_2$)$_2$Zn and (MeCH═

Table 4
The yield of homo diorganozincs and group transfer ability in mixed diorganozincs in their CuI catalyzed acylation with benzoyl chloride **10** in THF and THF:TMEDA (1:2) at room temperature.[a]

$$\underset{\textbf{6}}{R^1R^2Zn} + \underset{\textbf{10}}{PhCOCl} \xrightarrow[\text{THF or THF:TMEDA (1:2), 25 °C, 20 min}]{\text{10 mol\% CuI}} \underset{\textbf{11}}{R^1COPh} + \underset{\textbf{12}}{R^2COPh}$$

R^1, R^2 = *n*-Bu **a**, Ph **b**, $PhCH_2$ **c**, MeCH=CH **d**

Entry		R^1	R^2	Total yield, %[b,c] (**11**:**12**)[d]	
				In THF	In THF:TMEDA (1:2)
1	$\mathbf{6a_2}$	*n*-Bu	*n*-Bu	90	64
2	$\mathbf{6b_2}$	Ph	Ph	66	59
3	$\mathbf{6c_2}$	$PhCH_2$	$PhCH_2$	–[e]	–
4	$\mathbf{6d_2}$	MeCH═CH	MeCH═CH	–[e]	–
5	**6ab**	*n*-Bu	Ph	78 (60:40)	67 (7:93)
6	**6ac**	*n*-Bu	$PhCH_2$	91 (27:73)	54 (22:78)
7	**6ad**	*n*-Bu	MeCH═CH	–	–
8	**6bc**	Ph	$PhCH_2$	61 (21:79)	60 (58:42)
9	**6bd**	Ph	MeCH═CH	4 (100:0)	32 (100:0)
10	**6cd**	$PhCH_2$	MeCH═CH	20 (100:0)	16 (81:19)

[a] Molar ratio of **6**:**10** was used to be 2:1.
[b] GC yield for homo diorganozincs $\mathbf{6a_2}$–$\mathbf{d_2}$.
[c] The sum of GC yields of **11** and **12** for mixed diorganozincs.
[d] The ratio of GC yields of **11** and **12**.
[e] $\mathbf{6c_2}$ and $\mathbf{6d_2}$ could not be prepared in THF at room temperature.

CH)$_2$Zn, we did not compare the CuI catalyzed acylation yields of mixed benzyl and alkynyl zincs with their background yields. The group selectivity order of PhCH$_2$ > *n*-Bu > Ph > MeCH=CH in the acylation of catalytic iodozinc diorganocuprates, R^1R^2CuZnI in THF appears similar to the reported order for substitution of lithium diorganocuprates [9a,f,g]. However, further investigation will be necessary for the interpretation of the solvent effect in the group transfer ability of R^1 group depending on R^2 groups in the reactions of diorganocuprates, R^1R^2CuM (M = Li, MgX, ZnX).

3. Experimental

All reactions were carried out under a nitrogen atmosphere in oven dried glassware using standard syringe-septum cap techniques [41]. Quantitative GC analysis were performed on a Thermo-Finnigan gas chromatograph equipped with a ZB-5 capillary column packed with phenylsiloxane using standard technique. THF was distilled from sodium benzophenonedianion; alkyl bromides and bromobenzene were obtained commercially and purified using literature procedures. Mg turnings for Grignard reagents was used without further purification. $ZnCl_2$ (Aldrich) was dried under reduced pressure at 100 °C for 2 h and used as a THF solution. CuI was purified according to the literature procedure, dried under reduced pressure at 60–90 °C for at least 1 h and kept under nitrogen [42].

HMPA, NMP, DMPU, diglyme and TMEDA were distilled under reduced pressure and kept over molecular sieves under nitrogen. Toluene was distilled and kept over Na under nitrogen.

Grignard reagents, RMgBr (R = *n*-Bu, Ph, PhCH$_2$, MeCH=CH) were prepared in THF by standard methods and their concentrations were found by titration before use [43]. *n*-Bu$_2$CuMgBr **1a$_2$** and Ph$_2$CuMgBr **1b$_2$** were prepared by addition of 2 mol equiv of *n*-BuMgBr or PhMgBr, respectively to a suspension of 1 mol equiv of CuI in THF at −20 °C and stirring at that temperature for 15–30 min. For the preparation of *n*-BuPhCuMgBr **1ab**, first *n*-BuCu was prepared by addition of 1 mol equiv of *n*-BuMgBr to 1 mol equiv of CuI in THF at −20 °C and then by addition of 1 mol equiv of PhMgBr to freshly prepared *n*-BuCu at −20 °C and the solution was stirred at that temperature for 15–30 min.

n-Bu$_3$ZnMgBr **1a$_3$** and Ph$_3$ZnMgBr **1b$_3$** were prepared by dropwise addition of 3 mol equiv of *n*-BuMgBr or PhMgBr, respectively in THF to a solution of 1 mol equiv of $ZnCl_2$ in THF at −15 °C and stirring at that temperature for 15 min. For the preparation of *n*-BuPh$_2$ZnMgBr **1ab$_2$** and *n*-Bu$_2$PhZnMgBr **1a$_2$b**, freshly prepared Ph$_2$Zn and PhZnCl respectively were used. Ph$_2$Zn and PhZnCl were prepared in THF by dropwise addition of 2 mol equiv and 1 mol equiv of PhMgBr, respectively in THF, respectively to a solution of 1 mol equiv of $ZnCl_2$ in THF at −15 °C and stirring at that temperature for 5 min. *n*-BuPh$_2$ZnMgBr **1ab$_2$** was prepared by dropwise addition of 1 mol equiv of *n*-BuMgBr in THF to 1 mol equiv of Ph$_2$Zn in THF at −15 °C and stirring at that temperature for 15 min. *n*-Bu$_2$PhZnMgBr **1a$_2$b** was prepared by dropwise addition of 2 mol equiv of *n*-BuMgBr in THF to 1 mol equiv PhZnCl in THF at −15 °C and stirring at that temperature for 15 min.

n-BuPhZn **6ab** was prepared by addition of 1 equiv of *n*-BuMgBr and following 1 equiv of PhMgBr to 1 equiv of $ZnCl_2$ in THF at 0 °C at stirring at that temperature for 15 min. Mixed diorganozincs, R^1R^2Zn were prepared similarly using 1 equiv of R^1MgBr and 1 equiv of R^2MgBr.

For the coupling reactions of diorganocuprates, copper catalyzed coupling of triorganozincates and coupling of diorganozincs, CuI catalyzed coupling of triorganozincates is given below as a representative procedure:

To THF solution of zincate (1 mmol) prepared at −15 °C in a flame-dried and two-necked flask equipped with septum caps and a stirring bar, CuI catalyst (0.2 mmol, 0.0381 g) was added and the mixture was stirred at that temperature for 15 min. *n*-Pentyl bromide (**2**) (1 mmol, 0.12 ml) was added dropwise with a syringe. The mixture was allowed to warm to room temperature and stirred at required temperature for an appropriate time. After addition of an internal standard, the mixture was hydrolyzed with aqueous NH_3/NH_4Cl solution. The aqueous phase was extracted with ether and aliquots were analyzed by GC.

4. Conclusions

We carried out this work to show the dependence of group selectivity on the solvent in the reactions of diorganocuprates, either stoichiometric or catalytic and diorganozincs.

In conclusion, we have found that the use of a mixed cuprate R^1R^2CuMgBr (R^1 = *n*-alkyl, R^2 = aryl) instead of a homocuprate R^1_2CuMgBr for alkyl coupling in THF increases *n*-alkyl transfer and using a N- or O-donor cosolvent in 1:1 molar ratio or less does not change either the yield or the relative group transfer ability. However, in the alkylation of catalytic mixed cuprates derived from CuI catalyzed mixed zincates $R^1_2R^2$ZnMgBr and $R^1_2R^2$ZnMgBr (R^1 = *n*-alkyl, R^2 = aryl), group transfer ability depends on the solvation effect of N- and O-donor solvents and it can be controlled by using N-donor cosolvents. In the CuI catalyzed alkylation of *n*-Bu$_2$PhZnMgBr, mostly Ph group transfer takes place in THF whereas *n*-Bu group transfer takes place in the presence of NMP, DMPU or TMEDA. In the CuI catalyzed alkylation of *n*-BuPh$_2$ZnMgBr, the presence of NMP or DMPU increases *n*-butyl group transfer. It seems that the use of TMEDA as a cosolvent in THF appears promising in atom-economic *n*-alkyl–*n*-alkyl coupling using $R^1_2R^2$ZnMgBr (R^1 = *n*-alkyl, R^2 = aryl) reagents and *n*-alkyl bromides and coupling of $R^1_2R^2$ZnMgBr reagents in THF rather than homozincates, R^2_3ZnMgBr seems to be the best choice for aryl-*n*-alkyl coupling.

In the allylation of *n*-BuPhZn, selective Ph group transfer takes place without solvent effect. The results are briefly discussed in terms of solvation of the mixed diorganocuprates and use of coordinating solvents in atom-economic alkylation of *n*-alkyl(aryl) cuprates and allylation of (*n*-alkyl)arylzincs. Solvent effect on the group transfer ability in the acylation of catalytic mixed cuprates derived from R^1R^2Zn in the presence of CuI catalysis has been found to be dependent also on the R^1 and R^2 partnership. Further studies on the synthetic use of solvent control of group transfer selectivity in the reactions of mixed diorganocuprates and diorganozincs are underway.

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